5. ARES: The Mars Port (Phase 4)

Mars atmosphere is 0.6% of Earth's. The hard parts of the Earth machine are nearly free there.

The tube holds back 0.6% of the pressure the Earth tube fights; the plasma window's job is 99.4% done by the planet; exit heating is a fraction of the Earth case. Dust storms, not aerodynamics, are the main environmental design driver.

Element	Figure	Notes
Orbit track	~3.4 km/s → ~180 km @ 3.3 g	105-second run to low Mars orbit
Escape track	~5.0 km/s → ~390 km @ 3.3 g	155-second run; between Moon and Earth in scale, far easier than Earth in engineering
Energy	1.6 / 3.5 kWh per kg	orbit / escape
Staging	Phobos & Deimos	Catch points and propellant depots — the warehouse district; SEP tugs fly the cruise

Every Mars mission arrives needing return propellant. ISRU solves it on the surface (Sabatier methalox from CO_2 and water ice); ARES solves it in orbit — propellant rides to Phobos at electromagnetic cost and waits in a depot for any departing vehicle. Return cargo, manufactured exports, and samples ride the same track. **ARES makes Mars a place you can leave as easily as you arrive.**

> **The Moon as deep-space launch platform.** A Mars-transfer from the lunar surface costs ~3.0 km/s (≈1.25 kWh/kg) — roughly one-fifteenth the energy of the same trip from Earth's surface. Anything Mars-bound that can be built on, or staged through, the Moon ships from the lunar track: relay satellites, sensor networks, pre-positioned supplies. Starship carries crew and cargo that must arrive fresh; the mass drivers send everything else ahead, early and cheap. The Mars supply chain runs through the Moon, not around it. ARES is Phase 4: it inherits a mature vacuum-world kit (SELENE, Phase 1; ASTERIA, Phase 2–3) and proven atmosphere subsystems (Chimborazo, Phase 2–3).

6. The Build Order

The lightning flashes before the thunder is heard: SELENE launches silently in vacuum before BRONTE shakes the air on Earth.

Phase	System	What gets built	What it proves / earns
1	**SELENE PoC** (Moon)	1–3 km surface track, 50–100 g, south-pole ridge	2.9 km @ 100 g clears lunar escape; exports water/regolith/metal to cislunar buyers from day one; grows to 10–12 km (30 g) then ~50 km (3 g)
2–3	**Chimborazo** (Earth) + **ASTERIA**	14 km first build of through-mountain line (tube, plasma window, aero-shell) running in parallel with SELENE-class kits stamped on Phobos, Deimos, Ceres, the Belt	Chimborazo earns DoD test revenue and proves cost-per-km; ASTERIA deploys proven vacuum-world technology across airless bodies — both run concurrently, the core bet and the Earth demonstrator advance together
4	**ARES** (Mars)	~180 km track, featherweight tube	Mars becomes a port; the freight network closes
5	**BRONTE full** (Earth)	Extend Chimborazo west to ~38 km (Mach 12.6); beyond that, the 1,000 km-class 3 g injector if economics prove out	The endgame — built only on validated cost data

SELENE is ~300× shorter than any full Earth injector and skips every atmosphere subsystem. Chimborazo Phase 1 is ~70× shorter and proves the hard atmosphere parts before any large commitment. They de-risk in parallel.

7. How the Network Moves Mass

Three launchers, one supply chain. Once they exist, moving mass around the inner system mostly costs what running the launchers costs.

- **Earth sends what only Earth can build** — machines, electronics, medicine. The eventual full injector runs 80 t/day (29,000 t/yr); the Chimborazo demonstrator's job is to earn that machine the right to exist. (Rev 1 wrongly pinned the 80 t/day cadence on the demonstrator itself.)
- **The Moon sends raw mass** — propellant feedstock, shielding regolith, structural metal, fired at 2.4 km/s toward cislunar catch points. At full cadence the cislunar propellant supply stops being constrained by what Earth can lift.
- **Mars closes the return trip** — ISRU methalox, samples, and exports ride ARES to Phobos/Deimos depots on electricity.
- **SEP tugs connect the nodes** — weeks per leg, nearly propellant-free; chemical engines do only final descents. The launchers do all the heavy lifting off every surface.
- **Each node funds the next.** Rockets deliver SELENE's first coils (the last rocket buy the lunar build needs); SELENE exports fund Chimborazo; Chimborazo revenue and data fund ASTERIA; ASTERIA builds ARES. Every airless body added makes the whole network richer.

8. The Skyhook: How Cost Falls Toward $5/kg

A rotating tether catches pods and flings them higher. No propellant. More payload per launch.

A long tether rotates in low orbit; its bottom tip sweeps backward, moving slower than orbital velocity. A pod that arrives at the tip gets grabbed; half a rotation later it is released up high, moving faster — to the Moon, GTO, or wherever the mission calls. With a 1.5 km/s tip speed, the launcher needs 6.3 km/s instead of 7.8: ~35% less energy, a slower exit (less heating), and the kick stage shrinks from ~20% of pod mass to 2–5% for trim — the freed 15–18% becomes payload. That shift is where $5/kg comes from.

- **Keeping it up:** electrodynamic reboost (current through the tether against Earth's field — demonstrated physics, no propellant) plus momentum returned by catching descending capsules.
- **The lunar extension:** a cislunar rotovator catches SELENE pellets and redirects them to LLO, L1, or Earth-transfer; an Earth-orbit tether receives. Cargo crosses cislunar space without burning its own propellant at all.
- **Provenance:** momentum-exchange tethers date to the 1970s; the Boeing/Tethers Unlimited HASTOL study (NASA NIAC, 1999–2001) detailed a 600 km tether with 3.5 km/s tip speed catching Mach-10 aircraft-launched payloads. Materials and manufacturing at scale, not new physics.

9. If the Full Earth Injector Gets Built

Not the plan — ASTERIA is the plan. But if ASTERIA works, this is what becomes possible.

A full Earth injector — ultimately the 1,000 km-class, 3 g, 8 km/s human-rated machine — is the most consequential infrastructure ever proposed. The key point: if SELENE works in vacuum and Chimborazo validates the atmosphere subsystems and the cost-per-kilometer, the full injector stops being a physics question and becomes a construction question.

What $5–20/kg to orbit changes

- **Satellites:** launch stops dominating cost — a 2,000 kg satellite rides for ~$20K at $10/kg vs the $83–97M a GPS III mission actually paid. Constellations replace on maintenance schedules.
- **Stations:** ISS-class logistics (~$3.1B/yr, ~$1.8B of it transport) collapses ~300×; thousand-person stations become economically normal.
- **Orbital manufacturing:** ZBLAN fiber, crystals, alloys — Flawless Photonics drew ~12 km of ZBLAN on ISS in 2024 (1,141 m in one day vs a 25 m prior record); the candid caveat is that published proof of reproducibly superior fiber is still pending. The queue is real either way, and it is waiting on freight prices.
- **Deep space:** with cheap Earth-to-orbit plus lunar propellant, Mars missions stop being nation-sized budget exercises; the Belt becomes commercially reachable.

> **The strategic picture.** The operator of an 80 t/day injector controls the access ramp to space — 29,000 t/yr can replenish a damaged constellation in hours. The host nation stops paying for space access and starts charging for it. ASTERIA pays for itself; the Earth injector is what ASTERIA eventually makes possible.

10. Theoretical Scaling: Wider Bores and Multi-Barrel Batteries

Pure forward-looking theory. None of this is in the build plan — it is what the architecture could become if the demonstrator succeeds and the enabling technology keeps maturing.

Read this section as speculation, not specification. Everything below assumes the Phase-1 demonstrator has already worked and that several technologies have advanced past where they sit in 2026. These are not designs. They are the shape of the ceiling — included because the question "how far does this scale?" deserves an honest, numbered answer rather than hand-waving.

The one rule that governs all of it: bore sets the chunk size, power sets the tonnage. The energy to place a kilogram in orbit is fixed by physics — a wider tube does not make a tonne cheaper, it makes each shot bigger and rarer for the same megawatts. More annual tonnage comes only from more installed power. Bigger bores and more barrels change *how* you deliver the mass; the power plant decides *how much*.

Theoretical A — the Ø5 m bore

Payload scales with bore cross-section (diameter^2): widening the Ø3.2 m payload bay to Ø5 m is a **2.44× larger payload per shot**. The same exit velocity is required, so energy per shot scales with it.

What would have to advance first: the plasma window — already the program's hardest R&D at Ø4 m — would need to hold a still-larger aperture; the drive ring would need more REBCO at a better cost per kiloamp-metre than 2026 production; tunnelling would need bores beyond today's ~17.6 m TBM record; and trackside storage would need to swallow ~80 MWh per shot. Each is an extrapolation of a real 2026 capability, not a new invention — but all four have to arrive together.

Theoretical B — the multi-barrel battery

Instead of one fat tube, place several independent barrels side by side on one site — ~20 m apart through the rock (a rock pillar between Ø12 m bores, the way twin- and triple-bore road tunnels are already built), 5–10 m apart on open viaduct. They share one tunnel campaign, one substation, one cryoplant, one vacuum plant, one control centre, and — critically — **one storage farm fired round-robin:** while barrel 1 recharges, barrel 2 fires, then 3, so the grid sees a steady draw instead of three spikes and the site fires far more often than any single barrel.

What this does and does not buy: it is mostly engineering scale-up, not new physics — but it is gated on the demonstrator first proving the per-barrel cost, and on building power to match. Three barrels deliver ~3× the tonnage for under 2× the capital, because the expensive infrastructure (and the permitting fight) is paid once. It does not escape the power rule: three barrels at full cadence draw three times the megawatts.

How the numbers scale (theoretical)

Configuration	Payload / shot (cargo · bulk)	Energy / shot	Power feed	Annual / site (cargo · bulk)
Ø3.2 m single barrel (demonstrator-derived baseline)	4.5 t · 14 t	~33 MWh	~30 MW	~13,000 t · ~40,000 t
Ø5 m single barrel (theoretical)	~11 t · ~34 t	~80 MWh	~73 MW	~32,000 t · ~99,000 t

Triple Ø3.2 m battery (theoretical)	4.5 t · 14 t (×3, staggered)	~33 MWh ×3	~90 MW	~39,000 t · ~120,000 t

Cargo = 30 g, g-hardened satellites, with a kick stage to orbit. Bulk = 50–77 g inert mass (propellant, water, metal) into a depot. Annual figures assume ~8 effective shots/day per barrel at ~80% availability and that power is scaled to suit.

What it would take to reach 100,000 t/yr to LEO

Approach	Units needed	Notes
SpaceX Starship (reference)	~40 ships	at ~25 flights/ship/yr (Falcon-9-class reuse); ~1,000 flights/yr, ~2.7/day; ~4.6 Mt of methalox/yr
Ø3.2 m barrels	~8	eight sites, or fewer with batteries
Ø5 m barrels (theoretical)	~3	each fed ~73 MW
Triple Ø3.2 m batteries (theoretical)	~3 sites (9 barrels)	three mountains instead of eight

For scale: the entire world placed roughly 1,900 t in orbit in 2024. Every figure in this section is two orders of magnitude past that — and bounded, in the end, not by the launchers but by how much power and how many orbital catch-points can be built to absorb the mass.

PURE THEORETICAL SCALING — REQUIRES TECHNOLOGY MATURATION BEYOND 2026 — NOT A BUILD PLAN, NOT FOR CONSTRUCTION

11. Zero New Physics

Every subsystem has flown, fired, or been demonstrated. The opportunity is assembly, not invention.

Building block	Status — verified June 12, 2026
EMALS (US Navy)	Operational on USS Gerald R. Ford since 2017 incl. combat deployment; 484 MJ flywheel storage, 45 s recharge. CVN-79 (Kennedy) completed builder's trials Feb 2026, delivery ~Mar 2027. 2023 AFOSR-filed report recommends evolving EMALS for lunar launch.
Superconducting maglev	603 km/h crewed record (JR Central L0, 2015) still stands; Shanghai Transrapid commercial since 2003; Chūō Shinkansen now expected NET ~2034 after the Shizuoka settlement (early 2026).
REBCO superconductors	Tc ≈ 92 K. All-REBCO records: 26.86 T (ASIPP, 2024) → **28.2 T** (CHMFL, 2025); all-superconducting record **35.1 T** (ASIPP, Sept 2025); 45.5 T hybrid (NHMFL, 2019). CFS demonstrated 20 T SPARC-class coils (2021); SPARC assembly underway (first TF coil placed Jan 2026), first-plasma targeted 2026, Q>1 targeted 2027; ARC (Virginia) early 2030s. Jc > 1,000 A/mm^2 at 30 T *at 4.2 K*. Launch LSM needs only 5–10 T at the gap.
MHD plasma window	Invented by A. Hershcovitch (Brookhaven), patented 1995; holds ~9 atm; **demonstrated** for non-vacuum e-beam welding at cm-scale apertures. Scaling to the Ø4 m bore (2.8× baseline power) is the program's R&D centerpiece — Phase 1 tests it at full bore.

X-59 shock shaping	First flight Oct 28, 2025; **first supersonic flight June 5, 2026** (Mach 1.1); Mach 1.4 acoustic points next; XVS camera-vision cockpit flight-proven.
High-g electronics	Gun-launched guidance routinely qualified to 15,000–20,000 g (Excalibur-class). SELENE's 100 g cargo mode is conservative by two orders of magnitude.
EM launch-assist market	Auriga Space: $12.2M cumulative incl. AFWERX D2P2; hypersonic-test-as-revenue model proven in market; outdoor track (Thor) planned 2026.
Orbital manufacturing demand	Flawless Photonics: ~11.9 km ZBLAN drawn on ISS (Feb–Mar 2024; 1,141 m/day vs 25 m prior record); Luxembourg production expansion; published performance characterization still pending — flagged honestly.
Lunar mass-driver physics	O'Neill, NASA Ames studies 1976–77; Mass Driver One demo 1977. SJSU sizing study (2023): 25.4 kg pellets, 2.4 km/s, 10 s cadence, 8.7 MW polar solar, ~1.0 MA per bucket coil.
Lunar intent at scale	SpaceX lunar pivot + mass-driver statements (Feb–Mar 2026). Founder remarks, not a program of record — but the direction of the largest launch company on Earth.
Fusion pull (He-3 context)	NIF ignition Dec 2022 (3.15 MJ); best shot now **8.6 MJ, target gain >4** (Apr 2025), ignition repeated 9+ times. Commercial He-3 demand is already contracted: Interlune–Bluefors offtake (Sept 2025) for up to 10,000 L/yr from 2028, prospecting mission 2027.

12. The Hard Parts

A document about a project this large that doesn't name the real problems is salesmanship, not engineering.

- **Plasma window scaling.** Centimeter demos → Ø4.0 m bore at 2.8× baseline power. First R&D dollar; if it doesn't scale, the muzzle needs a different answer and we must learn that at demonstrator scale, cheaply.
- **Cost per kilometer is unknown.** The $20–250/kg projection lives or dies on vacuum-rated superconducting track cost. Nobody has published a credible figure. Chimborazo Phase 1 produces the first real one.
- **The El Arenal piers.** 110–590 m A-frames in a 0.4 g seismic zone — beyond Millau's 343 m record. Partial trenching, rock-buttressed spans, or accepting a lower exit are the studied outs.
- **The reserve.** Chimborazo sits inside a protected fauna reserve under a rights-of-nature constitution. The legal path may be harder than the rock. Stated plainly because pretending otherwise would be fatal later.
- **SELENE's first bill is a rocket bill.** First coils, power, and radiators ride to the Moon at today's prices; the starter track must fit one or two heavy-lift flights.
- **Catching is harder than throwing.** Until the first cislunar catch happens, export revenue is theoretical. Wide-cone tugs first, tethers later.
- **Polar ice uncertainty.** Shackleton-specific reserves are unproven (bounded ≤5–10 wt% in surveyed regolith). The early business plan must close on regolith, oxygen, and metal even in the lean-ice case. Chang'e-7 data lands within a year.
- **The treaty question.** The Outer Space Treaty is silent on commercial mass drivers; a kinetic launcher is inherently dual-use. That ambiguity buys scrutiny on one side and AFWERX funding on the other. EMALS and GPS both walked this road.

13. What KERAUNOS Unlocks

Orbital manufacturing

- **ZBLAN fiber** — up to 100× lower theoretical loss than silica; ~12 km drawn on ISS in 2024; customers queue on freight prices. KERAUNOS is the freight price. (Performance proof still pending publication — the bet is stated, not oversold.)
- **Orbital bioprinting** — organs that collapse under their own weight in 1 g; daily return pods turn it from stunt to logistics business.
- **Debris salvage** — thousands of tonnes of aerospace alloy already in orbit becomes ore when tug-hours are cheap.

Geopolitical position

- **The Space Panama Canal** — the host nation stops paying for access and starts charging for it.
- **Presence as a supply chain** — 80 t/day secures orbital slots and lunar territory as a logistics fact, not a flag.
- **The deterrence dividend** — constellation replenishment in hours, not months.

Fusion economy

- **The He-3 supply chain.** Lunar regolith holds an estimated (model-based) ~1M tonnes of He-3. D-He3 fusion is **low-neutron** (D-D side reactions keep it from being strictly aneutronic) — far less activation, direct conversion potential. At today's ~$20M/kg (~$20B/tonne) market price and SELENE's ~$5/kg lunar launch cost, delivery economics are absurdly favorable; Interlune's contracted offtakes (Bluefors, 2028–2037) prove demand precedes the fusion era. SELENE isn't just a launcher — it's the fuel logistics for the energy industry after next.
- **The energy loop.** Trackside storage doubles as grid-scale battery between launch windows; the same plant powers debris-sweeping lasers over the corridor. Nothing idles.

Verification note: all external claims in this revision were re-checked against primary reporting and literature on June 12, 2026; all physics re-derived from first principles the same day. Annex A (BRONTE Chimborazo Design Map) is bound after Section 4. The SELENE Design Map (KER-SEL-TRK-002) is issued as a companion sheet.